

Entity Relationship Diagram (ERD)

OptiCrop: Smart Agricultural Production Optimization Engine

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Team ID	
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Entity Relationship Diagram: The ERD below represents the data structure of the OptiCrop system. Since OptiCrop is a Machine Learning-based crop recommendation system using a CSV dataset (not a relational database), the ERD shows the single core entity — CROP_RECOMMENDATION — with all its attributes used to train the model and make predictions.

CROP_RECOMMENDATION

Key	Attribute Name	Data Type	Description
PK	id	INT	Unique record identifier (Primary Key)
-	nitrogen (N)	FLOAT	Nitrogen content in soil (kg/ha)
-	phosphorus (P)	FLOAT	Phosphorus content in soil (kg/ha)
-	potassium (K)	FLOAT	Potassium content in soil (kg/ha)
-	temperature	FLOAT	Average temperature (degrees Celsius)
-	humidity	FLOAT	Relative humidity (%)
-	ph	FLOAT	Soil pH value (0-14)
-	rainfall	FLOAT	Annual rainfall (mm)
-	label	VARCHAR	Recommended crop name — target variable (22 crop classes)

Legend

PK = Primary Key - = Regular Attribute (No foreign key — single entity model)

Note: OptiCrop uses a Machine Learning model (Random Forest Classifier) trained on the Crop Recommendation Dataset (2200 records, 22 crop classes). There is one entity (CROP_RECOMMENDATION) with no foreign keys since this is a single-table ML dataset, not a multi-table relational database.